

Tasks:

- 1. Install Nmap on your system.**
- 2. Verify installation:**
 - **nmap --version**
- 3. Scan your localhost:**
 - **nmap 127.0.0.1**
- 4. Scan a practice target:**
 - **nmap scanme.nmap.org**
- 5. Perform the following scans:**
 - **Basic Scan**
 - **Service Version Scan (-sV)**
 - **OS Detection (-O)**
 - **Fast Scan (-F)**
- 6. Identify:**
 - **Open Ports**
 - **Running Services**
 - **Service Versions**
 - **Operating System (if detected)**
- 7. Take screenshots of all scan results.**

Assignment:

Create a report containing:

- 1. What is Nmap?**
- 2. Why is Nmap used?**
- 3. What is a Port?**
- 4. Difference between Open, Closed and Filtered Ports.**
- 5. Screenshot of each scan.**
- 6. List all detected services and ports.**

7. Your observations.

Find the purpose of these ports:

- 21
- 22
- 25
- 53
- 80
- 110
- 143
- 443
- 3306
- 8080

Explain:

- Service Name
- Purpose
- Security Risks

Submission:

PDF or Word Document.

Objectives

- To understand the concept of network scanning using Nmap.
- To install and verify the Nmap tool.
- To perform different types of scans on target hosts.
- To identify open, closed, and filtered ports.
- To determine running services and their versions.
- To detect the operating system of the target machine.
- To analyze the obtained results and document observations.

Theory Questions:

What is Nmap?

Nmap (Network Mapper) is a free and open-source network scanning tool used for network discovery and security auditing. It helps administrators and security professionals identify hosts available on a network, discover open ports, detect running services, determine service versions, and identify the operating system of target systems.

Nmap was developed by Gordon Lyon (Fyodor) and has become one of the most widely used tools in cybersecurity.

Why is Nmap Used?

Nmap is used for various purposes, including:

- Network inventory and asset discovery.
- Detecting open ports on a host.
- Identifying services running on those ports.
- Determining software versions.
- Detecting operating systems.
- Monitoring host availability.
- Performing security assessments and vulnerability identification.

What is a Port?

A port is a logical communication endpoint used by network services to exchange data over a network. Each service running on a computer uses a specific port number.

For example:

- Web servers commonly use Port 80.
- Secure web servers use Port 443.

- SSH uses Port 22.

Ports allow multiple services to operate simultaneously using the same IP address.

Difference Between Open, Closed and Filtered Ports

Port State	Meaning
Open	An application is actively listening and accepting connections on the port.
Closed	The port is reachable but no service is currently listening on it.
Filtered	A firewall or filtering device is preventing Nmap from determining the exact state of the port.

Software Requirements

- Windows Operating System
- Nmap Version 7.99
- Zenmap GUI

Procedure

Step 1: Install Nmap

1. Download Nmap from the official website.
2. Run the installer.
3. Install Nmap along with Npcap.
4. Complete the installation process.

Step 2: Verify Installation

Open Command Prompt and execute:

```
nmap --version
```

Figure 1: Verification of Nmap Installation using Version Command

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\tusha>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 Npcap-1.87 nmap-
libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

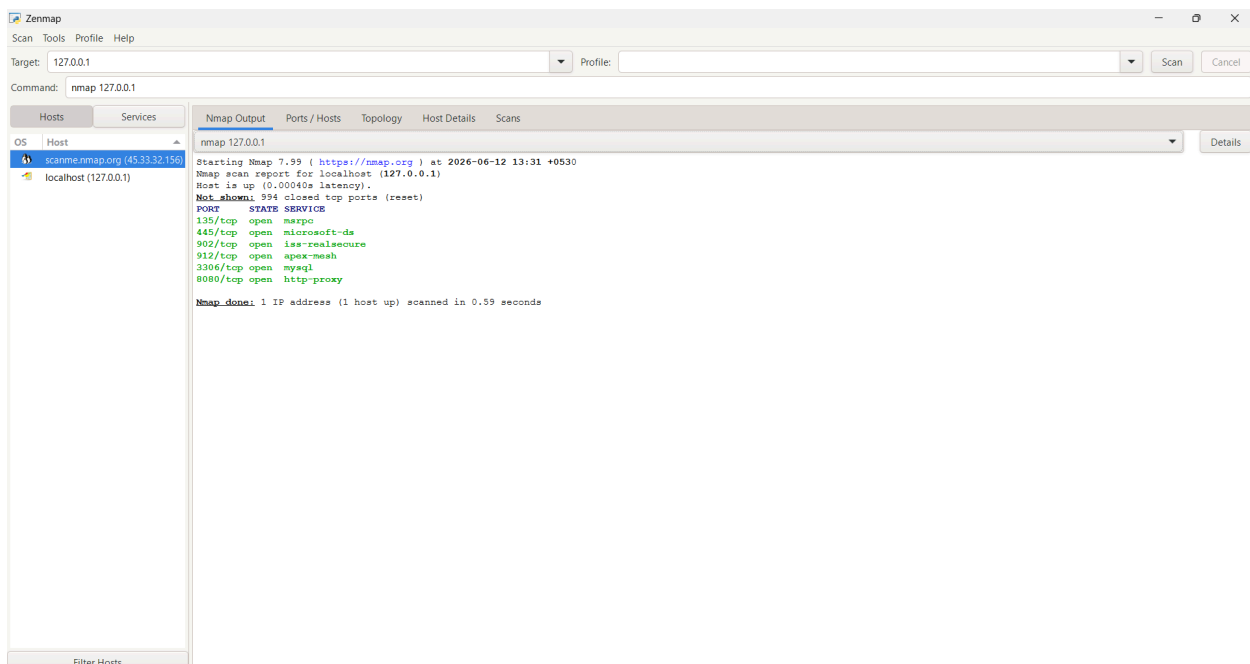
C:\Users\tusha>
```

Step 3: Scan Localhost

Execute:

nmap 127.0.0.1

Figure 2: Localhost Scan Result



Localhost Scan Results

Port	State	Service
135	Open	msrpc
445	Open	microsoft-ds
902	Open	iss-realsecure
912	Open	apex-mesh
3306	Open	mysql
8080	Open	http-proxy

Observation

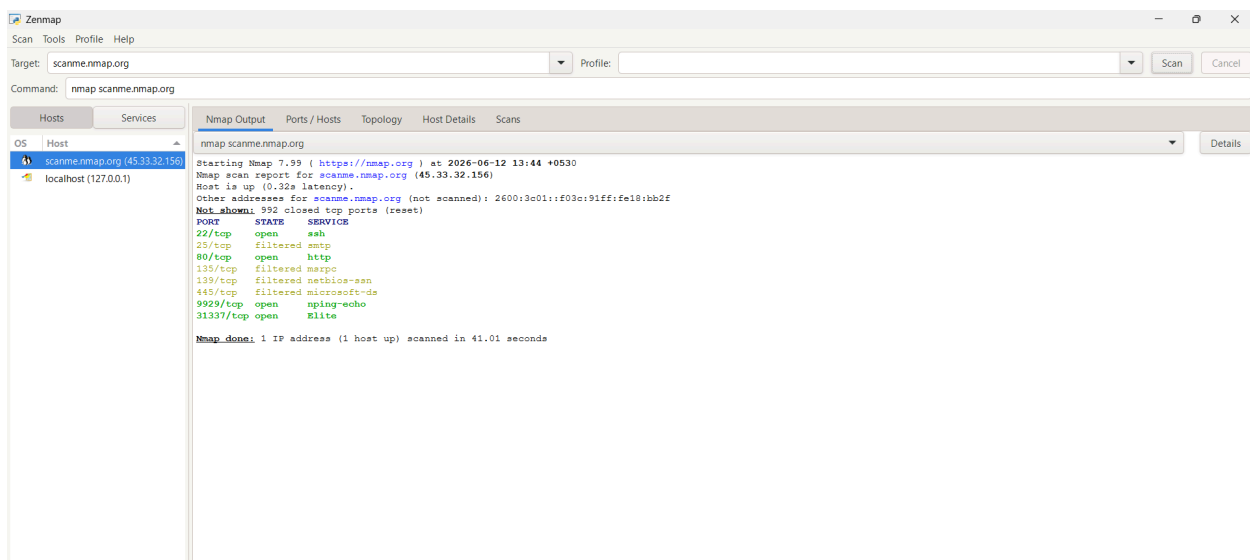
The localhost scan identified multiple services running on the system. Ports associated with Microsoft services, MySQL database service, and HTTP proxy services were found open.

Step 4: Basic Scan

Execute:

```
nmap scanme.nmap.org
```

Figure 3: Basic Scan of scanme.nmap.org



Basic Scan Results

Port	State	Service
22	Open	ssh
25	Filtered	smtp
80	Open	http
135	Filtered	msrpc
139	Filtered	netbios-ssn
445	Filtered	microsoft-ds
9929	Open	nping-echo
31337	Open	Elite

Observation

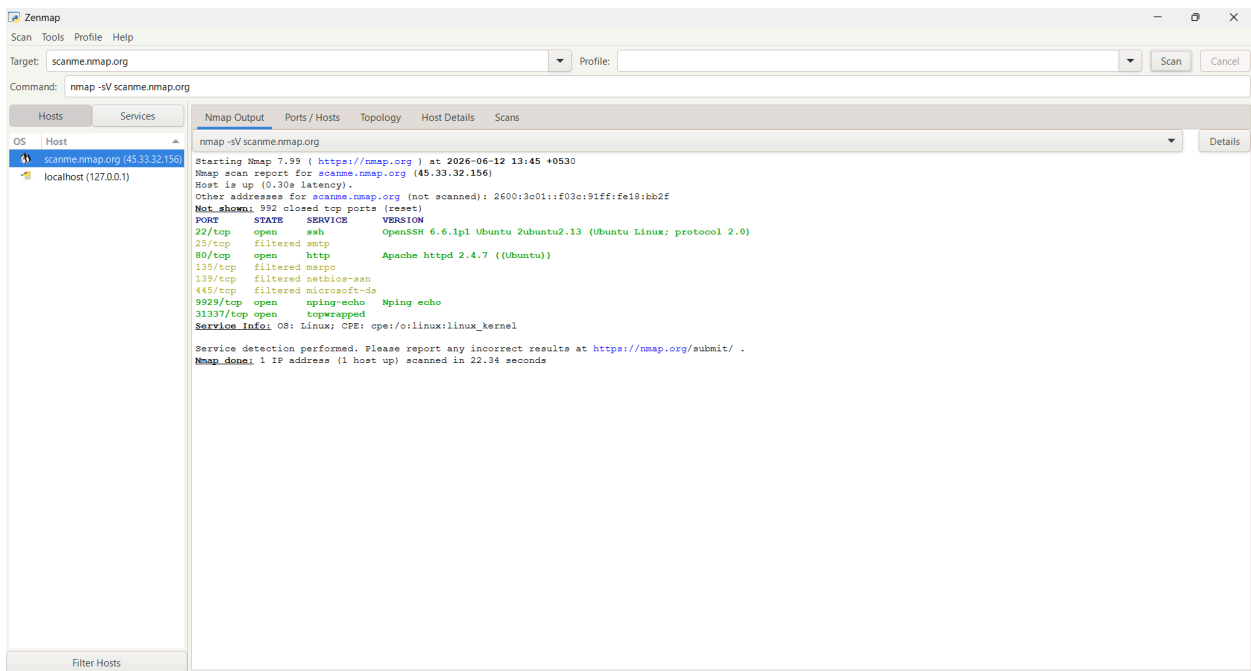
The target host had SSH and HTTP services accessible, while several ports were filtered by firewall rules.

Step 5: Service Version Detection

Execute:

```
nmap -sV scanme.nmap.org
```

Figure 4: Service Version Detection Scan



Service Version Results

Port	Service	Version
22	SSH	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
80	HTTP	Apache httpd 2.4.7 ((Ubuntu))
9929	nping-echo	Nping Echo
31337	tcpwrapped	Version not identified

Observation

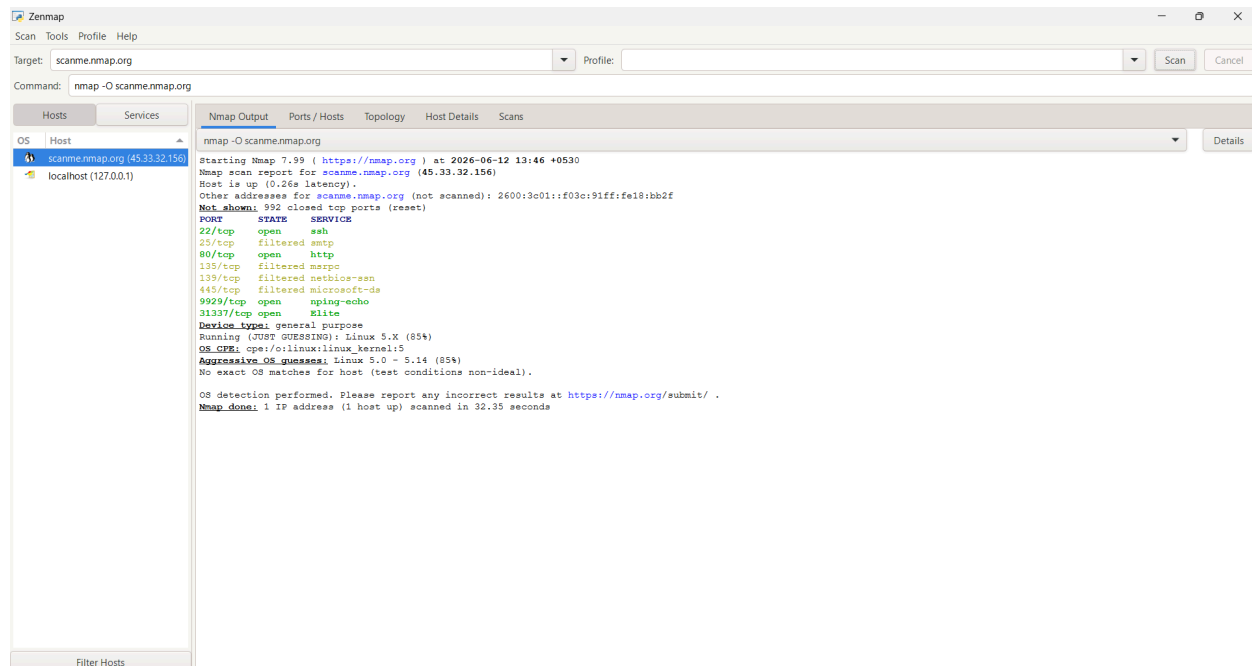
Nmap successfully identified software versions running on the open ports, providing detailed information useful during security assessments.

Step 6: Operating System Detection

Execute:

```
nmap -O scanme.nmap.org
```

Figure 5: Operating System Detection Scan



OS Detection Result

Parameter	Result
Device Type	General Purpose
Operating System	Linux
Running Guess	Linux 5.X
Accuracy	85%
Aggressive OS Guess	Linux 5.0 – 5.14

Observation

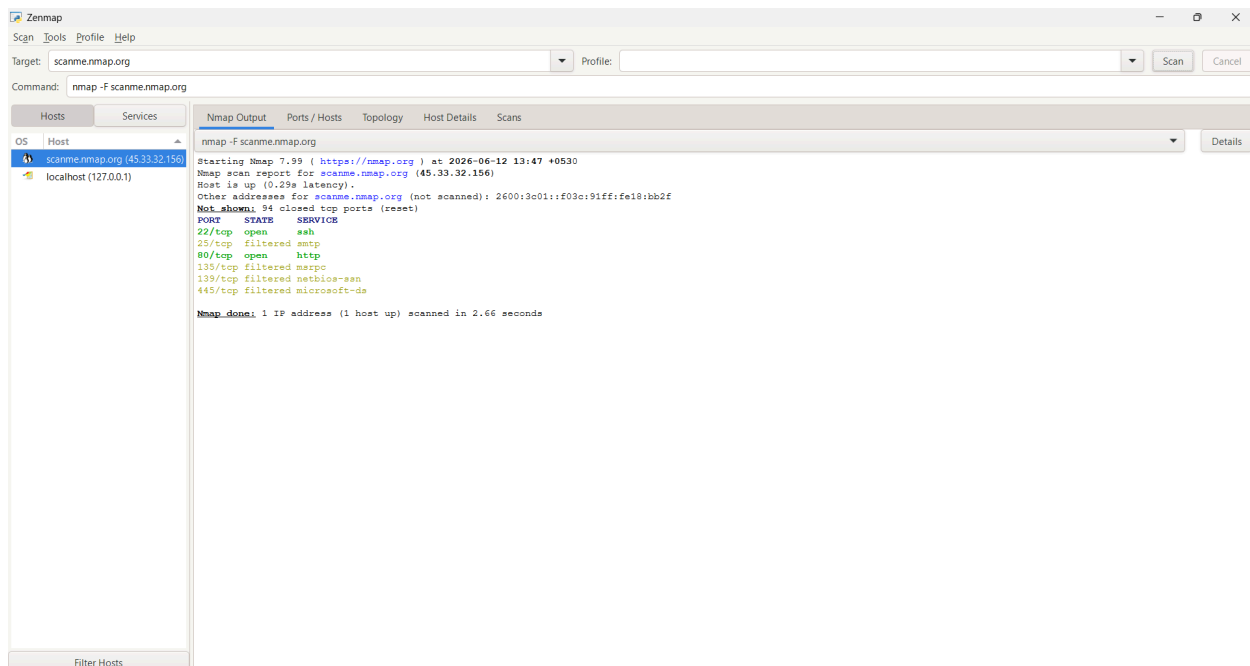
Nmap estimated that the target system was running a Linux-based operating system with an accuracy of approximately 85%.

Step 7: Fast Scan

Execute:

```
nmap -F scanme.nmap.org
```

Figure 6: Fast Scan Result



Fast Scan Results

Port	State	Service
22	Open	ssh
25	Filtered	smtp
80	Open	http

135	Filtered	msrpc
139	Filtered	netbios-ssn
445	Filtered	microsoft-ds

Observation

The fast scan reduced scan time by examining only the most commonly used ports while still identifying major services.

List of All Detected Services and Ports

Port	Service
22	SSH
25	SMTP
80	HTTP
135	MSRPC
139	NetBIOS Session Service
445	Microsoft-DS
902	ISS RealSecure
912	Apex Mesh
3306	MySQL
8080	HTTP Proxy
9929	Nping Echo
31337	Elite

Purpose of Common Ports and Security Risks

Port	Service Name	Purpose	Security Risks
21	FTP	File transfer between systems	Plain-text credentials can be intercepted
22	SSH	Secure remote administration	Brute-force login attacks
25	SMTP	Sending emails	Spam relay and email spoofing
53	DNS	Domain name resolution	DNS amplification attacks
80	HTTP	Web communication	Data transmitted without encryption
110	POP3	Email retrieval	Credentials transmitted insecurely
143	IMAP	Accessing emails from server	Unauthorized mailbox access
443	HTTPS	Secure web communication	SSL/TLS vulnerabilities if misconfigured
3306	MySQL	Database communication	Unauthorized database access
8080	HTTP Proxy/ Web Server	Alternative web services	Proxy misuse and web exploitation

Overall Observations

- Nmap version 7.99 was successfully installed and verified.
- Localhost scanning revealed several active services.
- The practice target had multiple open and filtered ports.
- Service version detection identified SSH and Apache versions.
- OS detection estimated the target to be running Linux.
- Fast scanning reduced scan duration while detecting major services.
- Nmap proved effective for network reconnaissance and security auditing.

Conclusion

The experiment was successfully completed using Nmap and Zenmap. Different scanning techniques such as basic scan, service version detection, operating system detection, and fast scanning were performed. Open ports, running services, and service versions were identified successfully. The practical enhanced understanding of network reconnaissance techniques and demonstrated the importance of port and service analysis in cybersecurity assessments.